

Samsung and LG agree to a truce

Both sides have agreed to throw out lawsuits pertaining to display patents against each other
<http://dgit.in/18mHmF5>

Moon younger by 100m years

New theory points out that the Moon may be a 100 million years younger than previous estimates
<http://dgit.in/16vrCwM>

From the labs

Flexpad: creating flexible projected interfaces on physical objects

Flexpad is a new project by the researchers from the Fluid Interfaces Group of the MIT Media Lab, that's evocative of MIT



An MRI scan being viewed on the FlexPad

Media Lab alumnus Pranav Mistry's 'Sixth Sense' projected interface project. The main point of focus of Flexpad is deformity- the projected interface will mould itself around physical objects such as paper or foam and will again re-mould itself when the shape of the object is changed. Users will also be able to deform objects to interact with the projected interface.

Flexpad differentiates itself from other flexible interface projects by ensuring that the deformities on the plane are captured in high detail and tracked in real-time. This is done with the help of a Kinect (the Xbox's camera accessory) and a projector, with the former helping to capture the depth and deformities of the object in front of it and the latter projecting the actual interface on the object's surface. Also, unlike previous projects that dealt with some level of projected flexible interfaces, Flexipad does not require the need for the object to have visual markers or textures and neither does it require that the object have some electronics embedded on it. The really neat part of the entire system is that it takes into account the fact that the user's hands or fingers will occupy a certain portion of the object's surface, and therefore, the Kinect camera is also used to identify and recognize the object as being separate from the user's appendages.

While Flexipad definitely sounds neat, it doesn't have the same kind of supreme mobility that Sixth Sense offered. According to the researchers, the current Flexipad setup involves a Kinect camera and a projector mounted to the ceiling above the

user which creates a 3D space measuring 110 x 55 x 43 cms within which the user can work with objects as varied as sheets of paper, foam or acrylic. In spite of this setup not being mobile, it will be interesting to see how the project evolves in the future.

You can check out a video of how Flexpad works here: <http://dgit.in/1563YaP>

MindRider: the bicycle helmet that can read your thoughts

This is a project that would be perfect for India and would reduce road-rage and spats that often lead to more chaos on the already anarchic situation on the roads. MindRider, developed by Arlene Ducao from the Information Ecology Group of the MIT Media Lab, is a bicycle helmet that flashes coloured lights determined by the wearer's mental state. It does this by reading the EEG (Electroencephalography) feedback (essentially, the electric signals transmitted by the brain along the scalp) from the wearer's brain and transmitting them to LED lights embedded along the surface of the helmet. If the lights are green, it means that the wearer is calm; red means that the wearer is anxious or drowsy while flashing red lights mean that panic and fear have come into the picture.

The MindRider prototype constructed by Ducao comprises of NeuroSky MindSet- a headset that performs EEG readings, an Arduino open-source single-board microcontroller, a Bluetooth radio so that the MindSet can communicate with the Arduino, an RGB LED light strip, transistors to control the light colours and a 9V battery. Ducao says that future versions of the MindRider could include better EEG readings, proximity sensors and visualisations that the cyclist/biker can see.

To get a better understanding of the MindRider, check out the video here: <http://dgit.in/188XDif>



Arlene Ducao wearing the MindRider

EyeRing: Giving the world the finger and getting a response

Unlike what the above title, conjured up by an author trying to act too clever, suggests, the EyeRing is a perfectly family-friendly device that is actually designed to help children and the disabled. Developed by Roy Shilkrot and Suranga Nanayakkara from the MIT Media Lab's Fluid Interfaces Group, EyeRing is a device worn around the ring-finger that gives the user information on what he/she is pointing at through either audio or visuals.

The EyeRing comprises of a micro-camera placed on a ring that can be slipped over a finger and it jumps to work when a user points the finger at an object and presses the button on the side of the ring. Doing so, causes the camera to shoot a picture or record a video and then transmits them automatically to a smartphone that has been paired to the EyeRing over Blue-



The EyeRing

tooth. The visual data is then translated into text or explained in detail by a custom app that's installed on the smartphone.

The folks at Engadget managed to get their hands on the EyeRing and while they did say that the device was buggy, they couldn't help but also recognise the potential it had. The two types of users that it would benefit most would be the visually impaired and children. In the case of the former, the EyeRing could help identify signs or currency and in the case of the latter, children could use the EyeRing to help in their learning process without the constant help of an adult.

Apart from pointing the finger in the air, the Eye-Ring also works if the user touches the surface of the object to be recognised.

Check out how the EyeRing works in this video: <http://dgit.in/18yMJ86>